

TAK Server - Data Package Creation from Scratch

TAK Server - Data Package Creation

- To create a data package from scratch, we need to first setup the appropriate file structure:
 - Navigate to the location you intend on storing the data packages and create a new folder named “Auto Enroll Data Package”
 - Within that “Auto Enroll Data Package” folder, create another folder named “MANIFEST”
 - Within the “MANFIEST” folder we are going to create a xml document named “manifest.xml”
 - This can be done by:
 - Right clicking within the folder, creating a new text document and then change its name
 - Open a new text editor, save the file with the correct name and in the intended location.

TAK Server - Data Package

Pre-requirements

- We need to get the immediate trust store for the server and the client certificate, if you aren't using auto enrollment.
 - Open a terminal in the server or ssh into the server:
 - If you need to create the client certificate (as the tak user):
 - **sudo su tak**
 - **cd /opt/tak/certs**
 - **./makeCert.sh client <name>**
 - Copy the files to the home directory (not as the tak user):
 - **cd ~**
 - **sudo cp -v /opt/tak/certs/files/<filename>.p12 .**

```
[takadmin@unknown08002763afdc ~]$ sudo cp -v /opt/tak/certs/files/DP-User.p12 .  
'/opt/tak/certs/files/DP-User.p12' -> './DP-User.p12'  
[takadmin@unknown08002763afdc ~]$ sudo cp -v /opt/tak/certs/files/truststore-TAK-ID-CA-01.p12 .  
'/opt/tak/certs/files/truststore-TAK-ID-CA-01.p12' -> './truststore-TAK-ID-CA-01.p12'  
[takadmin@unknown08002763afdc ~]$
```

TAK Server - Data Package

Pre-requirements

- Change the permissions back to the user:
 - **sudo chown -R takadmin:takadmin /home/takadmin**
- Open a terminal on your local machine and copy over the files using the following:
 - **scp <username>@<IP Address>:~/<filename>.p12 .**
 - These files will need to go into the “Auto Enroll Data Package” folder:
 - You can navigate the command line to the destination of the folder then download it there.
 - You can download the files then move/copy them over to the location.

```
C:\Users\640458\Downloads>scp takadmin@192.168.1.124:~/DP-User.p12 .
takadmin@192.168.1.124's password:
DP-User.p12
100% 4356 2.1MB/s 00:00

C:\Users\640458\Downloads>scp takadmin@192.168.1.124:~/truststore-TAK-ID-CA-01.p12 .
takadmin@192.168.1.124's password:
truststore-TAK-ID-CA-01.p12
100% 2080 1.0MB/s 00:00

C:\Users\640458\Downloads>
```

TAK Server - Data Package Creation

- Manifest break down:
 - Configuration:
 - **uid:** the uid for the data package helps identify it if the name is changed when sent from person to person. If the uid is not supplied, a hash is generated and used to determine if the file is unique.
 - **name:** the name of the data package for display to the end user.
 - **onReceiverDelete:** determines in the data package should be deleted as soon as the import is finished. This is ideal for staging devices.
 - Contents:
 - **Content:** the name of the file in the data package. If the ignore flag is set to true, the file is not passed into the import manager.
 - **Zip entry:** the standard Zip Entry path
 - **Ignore:** If true the zip entry is not extracted during import and is not listed in the mission package tool UI.
- Open the “manifest.xml” file with a text editor, copy the information on the next slide and change the highlighted parts to fit your deployment:

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```
<MissionPackageManifest version="2">
  <Configuration>
    <Parameter name="uid" value="a647112f-ce05-4312-a2b4-208d8d8a5fa8"/>
    <Parameter name="name" value="DP-TAK-Server.zip"/>
    <Parameter name="onReceiveDelete" value="true"/>
  </Configuration>
  <Contents>
    <Content ignore="false" zipEntry="config.pref"/>
    <Content ignore="false" zipEntry="truststore-TAK-ID-CA-01.p12"/>
    <Content ignore="false" zipEntry="DP-User.p12"/>
  </Contents>
</MissionPackageManifest>
```

- The grey highlighted section is the uid of the data package which can be used or changed to a unique one.
 - You can run **uuidgen** in the command line to generate one.
- The yellow highlighted section is the name you want for the data package.
- The green highlighted section is the name of the truststore file you download from the TAK Server.
- The blue highlighted section is the name of the client certificate you download from the TAK Server.
 - If you are using autoenrollment you do not need to include the client certificate.
- When done, save the changes.

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- Navigate back to the “Auto Enroll Data Package” folder.
 - Create a new file named “config.pref” and open it with a text editor and fill with the contents on the next slide. Do not forget to update the highlighted sections.

TAK Server - Data Package Creation - With Client Certificate

```
<?xml version='1.0' standalone='yes'?>
```

```
<preferences>
```

```
  <preference version="1" name="cot_streams">
```

```
    <entry key="count" class="class java.lang.Integer">1</entry>
```

```
    <entry key="description0" class="class java.lang.String">DP-TAK-Server</entry>
```

```
    <entry key="enabled0" class="class java.lang.Boolean">true</entry>
```

```
    <entry key="connectString0" class="class java.lang.String">192.168.1.124:8089:ssl</entry>
```

```
    <entry key="caLocation0" class="class java.lang.String">cert/truststore-TAK-ID-CA-01.p12</entry>
```

```
    <entry key="certificateLocation0" class="class java.lang.String">cert/DP-User.p12</entry>
```

```
    <entry key="clientPassword0" class="class java.lang.String">atakatak</entry>
```

```
    <entry key="caPassword0" class="class java.lang.String">atakatak</entry>
```

```
  </preference>
```

```
  <preference version="1" name="com.atakmap.app_preferences">
```

```
  </preference>
```

```
</preferences>
```

- The yellow section is the display name of the server on the client's end.
- The green section is the IP address of the server, alternatively you can use the public facing domain name if setup.
- The blue sections are the information for the CA certificate.
- The red sections are the information for the client certificate.

TAK Server - Data Package Creation - Autoenrollment

```
<?xml version='1.0' standalone='yes'?>
```

```
<preferences>
```

```
  <preference version="1" name="cot_streams">
```

```
    <entry key="count" class="class java.lang.Integer">1</entry>
```

```
    <entry key="description0" class="class java.lang.String">DP-TAK-Server</entry>
```

```
    <entry key="enabled0" class="class java.lang.Boolean">true</entry>
```

```
    <entry key="connectString0" class="class java.lang.String">192.168.1.124:8089:ssl</entry>
```

```
    <entry key="caLocation0" class="class java.lang.String">cert/truststore-TAK-ID-CA-01.p12</entry>
```

```
    <entry key="caPassword0" class="class java.lang.String">atakatak</entry>
```

```
    <entry key="enrollForCertificateWithTrust0" class="class java.lang.Boolean">true</entry>
```

```
    <entry key="useAuth0" class="class java.lang.Boolean">true</entry>
```

```
    <entry key="cacheCreds0" class="class java.lang.String">Cache credentials</entry>
```

```
  </preference>
```

```
  <preference version="1" name="com.atakmap.app_preferences">
```

```
</preference>
```

```
</preferences>
```

- The yellow section is the display name of the server on the client's end.
- The green section is the IP address of the server, alternatively you can use the public facing domain name if setup.
- The blue sections are the information for the CA certificate.
- The grey sections are optional additions but are recommend.

TAK Server - Data Package Creation

- From the “Auto Enroll Data Package” select the following content and zip them:

- **MANIFEST** directory

- **config.pref** file

- **CA.p12** file

- Example: **truststore-TAK-ID-CA-01.p12**

- **Client.p12** file

- If you are using autoenrollment you can skip this one.

- Example: **DP-User.p12**

- After zipping them change the name of the zip file to the intended one.

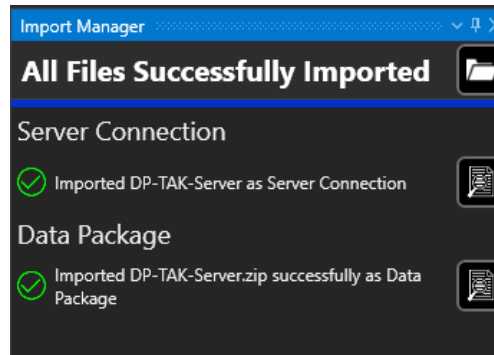
- This will need to be the same one you added in the “manifest.xml” file.

 MANIFEST		8/16/2024 8:57 AM	File folder	
 config.pref		8/16/2024 9:41 AM	PREF File	1 KB
 DP-User.p12		8/16/2024 9:22 AM	Personal Informati...	5 KB
 truststore-TAK-ID-CA-01.p12		8/16/2024 9:22 AM	Personal Informati...	3 KB

 DP-TAK-Server.zip

TAK Server - Deploying Data Package

- Either drag and drop the zip file into the TAK deployment or import the data package.
- If done correctly you will see the following after importing it:



- You can also verify the connection to the server was successful by locating the green cloud icon at the bottom right of the map.

TAK Server - Data Package - Manifest

- Manifest optional parameters:
 - **onReceiveDelete (boolean):** Omit or false to have mission packages persist on receiver, if true the zip will automatically be deleted after extraction on the receiver.
 - **onReceiveImport (boolean):** to have the mission package imported on receiver, if true, the mission package zip will be automatically imported. If false, the data will be extracted but not imported.
 - This is not processed for locally imported package but only for packages received over the network. (Sent by another ATAK user)
 - **onReceiveAction (string):** action which ATAK uses to broadcast an intent upon receiving a mission package.
 - ATAK includes the file transfer details and mission package manifest as intent extras. This allows tools to include custom configuration parameters which are delivered to the receiver component.
 - Note: if onReceiverDelete is specified, the mission package is deleted prior to instantiating and broadcasting this intent.

Resources

- Data package tool integration:

<https://wiki.tak.gov/display/DEV/Data+Package+Format?preview=/8323647/32048030/Data%20Package%20Tool%20Integration.pdf>